

Digital Agriculture Gap Report: West Africa and Burkina Faso

Bridging the Distance Between Tool Design and Farmer Reality

Prepared for: Agricultural development practitioners, digital innovation teams, extension officers, and agripreneurs designing technology for smallholder farmers in West Africa and Burkina Faso.

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Executive Summary

Digital agriculture tools and AI-powered advisory services hold genuine promise for African farmers facing climate shocks, market barriers, and resource constraints. Yet between the marketing claims and on-farm reality lies a substantial gap. Farmers in West Africa and Burkina Faso report that tools marketed as "intelligent," "adaptive," and "locally tailored" frequently deliver generic advice, fail in low-connectivity settings, operate in foreign languages, and often do not account for the actual crops, soils, and seasonal dynamics farmers know.

This report documents five critical gaps between aspiration and implementation, grounded in recent discussions (2020–2025) with farmers, extension workers, agripreneurs, and development practitioners, as well as analysis of institutional reports from FAO, World Bank, CGIAR, IFAD, and digital agriculture case studies. The report then outlines actionable recommendations for tool developers, extension services, and farmers on how to choose, adapt, and scale digital agriculture effectively.[1][2][3][4][5]

1. The Five Critical Gaps

Gap 1: Weather Information Is Unreliable or Inaccessible

The Problem

Farmers cite unreliable or poorly localized weather information as their top digital agriculture frustration.[2][4] Global weather models and continental-scale forecasting systems often miss the hyperlocal rainfall patterns, micro-climates, and seasonal quirks that determine whether a planting date will succeed or fail in a specific district or even a specific village.[6][7] Seasonal forecasts for the West African Sahel show persistent

underperformance, with models exhibiting dry bias and poor calibration at critical mid-range quantiles (rainfall amounts between 20–90 percent of normal).[7]

In addition, more than half of Africa's radar stations lack the capacity to produce accurate enough data for medium-range weather predictions.[8] Tools that claim to "predict the next 10 days of rainfall" often perform no better than a local elder's historical observation, especially during irregular or extreme seasons—exactly when farmers need the most guidance.[8]

Why This Matters

Farmers in rain-fed agricultural systems make planting decisions around three weeks before the onset of rains. If the weather forecast is wrong by a week or more, or if it misses localized thunderstorms or dry spells, farmers commit resources (labor, seed, fertilizer) to a crop that may fail. In Burkina Faso and the Sahel, where soils are fragile and input costs are high relative to income, this risk deters adoption of any new technology—digital or otherwise.[9]

Current State of Practice

- Some platforms (e.g., M-Shamba in Kenya, platforms in Senegal) deliver weather alerts via SMS or IVR (Interactive Voice Response), which works on feature phones. However, the accuracy of the underlying forecast remains modest.[10]
- National meteorology agencies (e.g., ANAM in Burkina Faso) provide agro-meteorological bulletins intended to guide farming decisions, yet their dissemination is limited and often in French or French-dominant formats, missing farmers in outlying areas who speak Moore, Dioula, Fulfuldé, or other local languages.[3]
- Farmers in Burkina Faso report that radio programs mentioning agriculture often reflect limited agricultural extension knowledge, and WhatsApp weather links rely on internet that half the rural population lacks.[3]

What a Mature Solution Would Require

1. Collaboration with national meteorology agencies and local agricultural research to validate forecasts at district and sub-district scales, not just national averages.
2. Integration with user feedback loops: if a forecast is wrong, the system learns why and adjusts parameterization for next time.
3. Transparent communication of forecast uncertainty ("60% confidence rain will fall between 15–50 mm on day 7") rather than point predictions.
4. Dissemination in local languages via SMS, IVR, and voice calls, not just web or app interfaces.

Gap 2: Crop-Specific and Soil-Specific Advice Is Generic or Absent

The Problem

Advisory platforms marketed as "AI-driven" and "personalized" frequently provide blanket recommendations (e.g., "apply 100 kg/ha of NPK fertilizer") that do not account for local crop varieties, soil types, prior yields, or farmer-specific constraints.[1][4][11] West African farmers cultivate dozens of crops and hundreds of local varieties: sorghum, millet, maize, cowpea, groundnut, cassava, rice, shea, cotton, and many others. Each has distinct

phenology, water demand, pest profile, and soil nutrient requirement. Few digital tools can model more than two to three crops with any agronomic depth.[1]

Moreover, soil heterogeneity within a single farm or village is extreme. Farmers know this: they speak of "black soil," "sandy soil," "stony soil," with intuitive understanding of which crops suit which patches. Digital tools rarely incorporate this hyper-local soil knowledge. Instead, they rely on coarse global soil maps or ask farmers to purchase soil testing—a service many cannot afford, and even fewer trust.[3][4][11]

Why This Matters

When a tool recommends a crop or practice that does not match a farmer's soils or varieties, the farmer loses trust in all digital advice. This is not abstract risk aversion; it is rational skepticism. A farmer who loses money following a generic recommendation will not retry digital advisory, and word spreads quickly in communities.[11]

Additionally, Sahel soils are often low in organic matter, prone to crusting, and highly sensitive to input timing. A recommendation that works in Kenya (higher rainfall, deeper soils) will fail in Burkina Faso. Yet many platforms are built on datasets from East or Southern Africa and scaled without regional recalibration.[1][4]

Current State of Practice

- A 2023 Burkina Faso case study found that the Ministry of Agriculture and Animal Resources (MARAH) has developed agro-meteorological advisory services via Viamo's IVR system (short number 321) that provide information in local languages, but the content is seasonal and general (e.g., "This is the planting season for sorghum"), not tailored to individual farmer plots or varieties.[3]
- Platforms like Complete Farmer (Ghana) and M-Shamba (Kenya) integrate expert agronomic input via call centers, but scale is limited to a few crops and regions.[1][4]
- Soil-mapping initiatives (e.g., Zenvus in Nigeria, catering to precision agriculture) exist, but cost is prohibitive for smallholders. These are typically used by larger-scale or contract farming operations.[1]

What a Mature Solution Would Require

1. Farmer-validated crop and variety profiles built from on-farm trial data in each agroecological zone (Sahel, Sudan savanna, guinea savanna, for Burkina Faso).
 2. Soil testing kits that are affordable and integrated with advisory (e.g., farmer takes a simple soil color and texture photo; AI + local extension interpret it).
 3. Agronomy rules that encode the decision logic of respected local farmers and research agronomists, with transparency: "For this soil type and rainfall pattern, sorghum variety X yields Y% better than variety Z."
 4. Multi-year feedback: if advice fails, the system documents why and updates the model.
 5. Modular design so that advice for a specific crop (e.g., cowpea in sandy soil) can be added and validated independently without rebuilding the entire platform.
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Gap 3: Pest and Disease Diagnosis Is Unreliable, Lacking Local Crops, or Insufficient Without Expert Follow-Up

The Problem

AI-powered disease diagnosis apps (e.g., PlantVillage, Plantix) use image recognition to identify crop diseases and pests. In principle, this is transformative: a farmer photographs a sick leaf, the app diagnoses the problem and recommends treatment. In practice, several challenges persist.[12][13]

First, most image-recognition models are trained primarily on images of diseases in temperate crops or large-scale cash crops (maize, wheat, rice, tomato). Local Sahel crops—millet, sorghum, cowpea, bambara groundnut, fonio, shea, baobab—are vastly underrepresented or absent from training datasets.[12][13] A farmer with sick millet plants will get a "no diagnosis found" or a misidentification, destroying trust.[13]

Second, the app diagnoses the disease but does not account for context. In a farmer's field, a disease may be present because of prior flooding, soil compaction, or lack of rotation—issues the photo alone cannot reveal. The app recommends fungicide; the farmer sprays, but the disease recurs because the root cause (waterlogging) was never addressed.[4][12]

Third, even where diagnosis is accurate, recommended treatments (chemical names, dosages) may not be available locally, or the farmer may lack the cash to buy them at the moment treatment is needed.[1][4][11]

Why This Matters

Pests and diseases are among the top five reasons farmers lose crops in West Africa. Rapid, accurate diagnosis could save yield; wrong or incomplete advice squanders the farmer's response window. The result: farmers revert to intuition, neighbor advice, or (more commonly) accept the loss, then become reluctant to try any new advisory tool.[1][4][12][13]

Current State of Practice

- PlantVillage Nuru (CGIAR) and other initiatives have built databases of disease and pest images for major African crops, but these remain geographically and crop-specific biased.[12]
- Some NGOs and extension services combine photo-based apps with human expert verification via WhatsApp or a call center (e.g., CABI's Plantwise Knowledge Bank), which raises accuracy but reduces scalability and introduces human bias.[1][4]
- FarmView (Carnegie Mellon, deployed in Mozambique) includes speech-to-text so farmers can describe the problem if photos are unclear, but such services are rare and not scaled to Burkina Faso.[1]

What a Mature Solution Would Require

1. Crowdsourced imagery: farmers photograph their own diseased crops with the app; human experts (extension agents, researchers, experienced farmers) verify the diagnosis and suggested treatment; the app learns from each case.
2. Treatment recommendations tied to local availability and affordability: "For this disease, recommended treatments are: (a) sulfur dust (widely available, affordable);

- (b) insecticidal soap (harder to find locally); (c) neem-based spray (common practice locally)."
3. Agronomic context integration: if a photo shows a disease that typically occurs after flooding, the app asks "Has this field been waterlogged recently?" and suggests drainage or intercropping as preventive measures.
 4. Sahel-crop datasets: active collaboration with ICRISAT, national research programs, and farmer groups to photograph and label millet, sorghum, cowpea, groundnut, and other crops at all growth stages and disease states.
 5. Connection to input supply: the app integrates with local agrodealer networks so treatment recommendations link to actual products available within 5 km, at known prices.
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Gap 4: Tools Are Not Usable on Low-End Phones or in Offline Mode, Excluding Most Rural Farmers

The Problem

Smartphones in Burkina Faso are concentrated among urban youth and the middle class. Most rural farmers rely on basic feature phones (2G/3G, no app store, limited processing power) or older Android smartphones with < 2 GB RAM. Yet many digital agriculture tools require smartphones with high specifications, stable 4G or 3G connectivity, and frequent internet access to function.[3][4][11]

Additionally, rural connectivity is intermittent. A farmer in a northern region may have signal in the morning but none by afternoon, or signal only on hilltops. Tools that require constant connection fail when farmers are in fields during the most critical decision moments (early morning, during storms, at field edges).[3][4][11]

Many platforms lack offline-first architecture. This means they cache nothing locally; they constantly fetch from the cloud. In low-bandwidth zones, users see spinning loading icons, timeouts, and frustration. If the user pays per megabyte for data (as is common in rural Africa), the cost becomes prohibitive.[4][11]

Why This Matters

If a tool does not work on the phone a farmer actually owns, or it costs too much data to use, adoption will remain near zero regardless of the quality of the advice. Aspirational design ("farmers will upgrade to smartphones") ignores the economic reality: a basic feature phone costs \$5–15 used; a modern smartphone costs \$100+, representing a month's income.[3][4][11]

Additionally, women, youth, and IDPs (internally displaced persons) in conflict-affected regions often lack reliable phone ownership or data allowances, so tools that demand connectivity inadvertently exclude the most vulnerable.[3][4][11]

Current State of Practice

- The IVR service 321 (Viamo, Burkina Faso) works on any phone (feature phone or smartphone) via voice call, making it one of the few truly accessible platforms in the country.[3]
- M-Shamba uses SMS and IVR in addition to apps, achieving broader reach than app-only competitors.[1][4]

- Most platforms built by startups or tech teams in capital cities default to app-only, inadvertently excluding rural users.[1][4][11]
- A 2023 Rethink Priorities report on weather forecasting noted that global weather models' accuracy remains poor in low-income regions; local offline calculation and caching could improve usability, but this is rarely implemented.[8]

What a Mature Solution Would Require

1. Offline-first architecture: the app downloads essential data (planting calendar, pest/disease ID database, input prices) weekly via WiFi or when signal is available; all core features work without connectivity.
2. SMS and voice IVR fallback: if the app does not load, critical advisory (e.g., "Rain forecast: plant tomorrow") goes via SMS or a 30-second voice call.
3. Low-specification design: the app runs on phones with 512 MB–1 GB RAM and Android 5.0+, not just flagship devices.
4. Data efficiency: the app uses < 5 MB per month of data; compression, chunking, and selective sync reduce bandwidth demand.
5. Local content hosting: instead of fetching from distant servers, district extension offices or mobile operators cache locally relevant content (local prices, regional pests, proven practices).

Gap 5: Language Localization Is Minimal; Tools Serve Elite/Educated Users, Not Majority Farmers

The Problem

Most digital agriculture tools are designed in English or French, with limited or no translation to local languages. Burkina Faso has five national languages (Moore, Dioula, Fulfuldé, Lélé, and French) plus dozens of minority languages. A farmer who does not speak French or English fluently will struggle with interfaces, instructions, and agronomic terms in a foreign language. This is not a matter of "digital literacy"; it is linguistic exclusion.[3][4][11]

Even where tools exist in local languages (e.g., the Viamo IVR service in Burkina Faso), the content is often recorded by non-farmers or urban staff, with terminology or pacing that does not match how rural farmers discuss crops and problems. A farmer hearing a rapid-fire technical explanation on a crackling phone line may not retain actionable knowledge.[3][4][11]

Additionally, many platforms are designed with the assumption that farmers are literate. Even where literacy rates have improved, many rural farmers, especially women and elders, prefer oral communication. Platforms that rely on text-based interfaces, written instructions, or assumption of formal education exclude large populations.[1][4][11]

Why This Matters

If a farmer hears advisory in a language they do not speak fluently, they will not trust the advice, will not remember the details, and will not adopt the recommended practice. Language is not a "nice-to-have" for inclusion; it is a core requirement for effectiveness.[3][4][11]

Moreover, solutions marketed as "using AI for local languages" (e.g., "trained on 50,000 Swahili sentences") rarely account for agronomic dialect: the words farmers use to describe a crop, a pest, or a soil condition are often not the formal academic terms. A farmer says "millet is yellow" (a sign of nitrogen deficiency); the AI expects "chlorosis" or "nitrogen deficiency." The mismatch causes failure.[1][4][11]

Current State of Practice

- Viamo's 321 IVR in Burkina Faso operates in five national languages (Moore, Dioula, Fulfuldé, Lélé, French), achieving ~130,000–150,000 calls per month. This is one of the strongest examples of language-localized advisory in the region.[3]
- Crop2Cash in Nigeria trains its AI models in seven local languages, including Yoruba and Igbo, and has seen success with smallholder uptake. However, scaling this approach requires significant investment per language and region.[1][4]
- Most web-based platforms (e.g., from startups, tech hubs) default to English or French, with token translations if any.[1][4][11]
- Radio remains popular for agricultural extension, but presenter expertise in farming is often limited. Community radio can be a vector for local-language advisory if content is co-produced with extension agents and farmers.[3][4][11]

What a Mature Solution Would Require

1. Participatory content design: farmers, extension agents, and researchers co-create advisory content in local languages, recording farmers' own voices and terminology.
2. Voice-first interfaces: the platform prioritizes audio (voice calls, voice messages, audio snippets) over text, with local-language speakers narrating and explaining recommendations.
3. Community radio integration: short (2–3 minute) audio segments broadcast on community radio weekly, covering seasonal pests, planting dates, input sourcing, and market information in local languages.
4. Farmer feedback loops: if a recorded message is unclear, farmers call a toll-free number and report problems; content is re-recorded and re-tested.
5. Collaboration with local language experts and linguists to ensure terminology matches how farmers actually speak about crops and problems.

2. The Gap Between Farmer-Perceived Problems and Tool Designers' Assumptions

A recurring theme across discussions with West African farmers, extension workers, and agripreneurs is that tools are designed with assumptions that do not match on-farm reality. [1][4][11]

Designer Assumption	Farmer Reality	Impact
Farmers want to optimize yield at all costs.	Farmers prioritize food security and stable income; risk aversion is rational, not a barrier to overcome.	Tools framed as "maximize yield 20%" may not fit farmer priorities if they increase variability or input cost.
Better data leads to better decisions.	Farmers already make decisions based on deep local knowledge; they want to integrate new information, not replace intuition.	Tools that ignore farmer knowledge or present contradictory advice are distrusted.
Digital tools are cheaper than extension agents.	Digital tools are only useful if they answer questions extension agents currently cannot address or answer faster/more frequently.	Farmers want extension agents + digital tools, not one or the other.
Farmers will upgrade their phones for an app.	Farmers invest in what is economically rational for them. A \$100 smartphone is not justified for one feature.	Tools must work on the phone farmers already own and can afford.
One recommendation fits all farmers in a region.	Farmers are highly differentiated by wealth, land size, family labor, soil, water access, and priorities. Generic advice fails frequently.	Segmentation and adaptive recommendations are essential, not nice-to-have.
Weather forecasts are useful if they are "more accurate than random."	Farmers need forecasts accurate to ± 5 days and localized to their village; coarse accuracy at national scale is less useful.	Seasonal forecasts (3 months ahead) are useful for planning; daily forecasts matter for immediate decisions.

3. Opportunities for Improvement: Actionable Recommendations

For Tool Developers and Tech Teams

1. **Start with farmer co-design, not technology.** Before coding, spend 2–3 months in the target communities observing decision-making, listening to problems, and prototyping low-tech solutions (paper cards, radio, SMS). Only then build digital tools.
2. **Build offline-first and SMS-first, not app-first.** Assume users have 2G and intermittent signal. Design so that the app works without connectivity, and critical information flows via SMS or voice calls.
3. **Localize to real farmer language, not formal academic terms.** Record advisory in farmers' own voices using their terminology. Test with target farmers; if they cannot understand a message, re-record it.
4. **Integrate with existing extension systems, not replace them.** Digital tools should amplify extension agents' reach (e.g., SMS updates that agents reference in group meetings), not displace them. Agents need training on the tool and buy-in that it is a complement.
5. **Build feedback loops and iterate quickly.** Deploy early and with small cohorts; measure adoption, trust, and behavior change; adjust content and design rapidly. A tool that works for 10 farmers can be scaled; a tool that does not earn trust from the first 10 will fail.
6. **Partner with national research and extension institutions from day one.** Content credibility depends on endorsement by respected agronomists and extension services. Government endorsement also signals sustainability and scale.
7. **Monetize responsibly.** Free tools to farmers; charge government agencies, NGOs, or private companies that benefit from aggregate insights or market linkages. Do not put the cost on the farmer.

For Extension Services and Government

1. **Map the digital landscape locally.** Conduct an audit of what tools already exist in your region, what works, what fails, and why. Do not assume new tools are better; often, combinations of existing tools (radio + SMS + extension agent visits) are most effective.
2. **Train extension agents on digital advisory platforms.** If a government or NGO is promoting a digital tool, fund training for the extension workers who will support it. An untrained extension worker cannot explain the tool to farmers.
3. **Invest in agro-meteorological content and dissemination.** National meteorology agencies (ANAM in Burkina Faso, ASECNA in Sahel) produce valuable forecasts but disseminate them poorly. Invest in SMS/IVR translation to local languages and regular farmer feedback on forecast accuracy.
4. **Support soil testing and agronomic trial networks.** Encourage farmers and research stations to conduct on-farm trials of varieties and practices in your zone. Farmers are more likely to trust advisory based on trials they see (neighboring farms) than distant research results.
5. **Regulate and accredit digital agriculture services.** If companies are selling advisory services to farmers, ensure a basic standard: accuracy claims are disclosed, feedback mechanisms exist, and farmers have recourse if advice fails.

6. **Create incentives for tool developers to localize.** Public procurement or subsidized licensing can make localization economically viable for small teams. A tool that works in English but not Moore is a failed investment if the target farmer speaks Moore.
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For Farmers and Farmer Groups

1. **Approach digital tools with healthy skepticism.** Before adopting a new tool, ask: Is this answering a real problem I have? Can I understand how it works? Is it affordable in time and money? Can I revert to my old method if the tool fails?
 2. **Demand local relevance.** If a tool does not work on your phone, or does not mention the crops you grow, or is in a language you do not speak, do not adopt it. Tell the tool provider why it failed.
 3. **Test on a small plot first.** Do not change your entire farming practice based on advisory from a tool you have never used. Test on a small area (even 10 m²) for one season.
 4. **Combine digital advice with human advice.** Talk to your neighbors, your extension agent, and your farmer association. If digital advice contradicts trusted local knowledge, dig deeper before adopting.
 5. **Share feedback and failures.** If a tool's advice fails, report it. If it works, recommend it. Word-of-mouth is how farmers decide; tool providers cannot improve if they do not hear about failures.
 6. **Organize into groups to reduce cost and increase influence.** A farmer association has more bargaining power with tool providers, can negotiate better pricing, and can collectively demand localization and support.
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4. Integrating Digital Tools with Traditional Extension: A Model

The most successful agricultural advisory systems in West Africa are hybrid:

- **National and regional meteorology:** official forecasts, validated but coarse.
- **Extension agents:** trained, trusted, know farmers personally, but limited reach and frequency.
- **Digital tools (SMS, IVR, radio):** high reach, low cost, but impersonal and sometimes inaccurate.
- **Farmer networks:** word-of-mouth, peer learning, most trusted, but narrow geographic reach.

Effective systems layer these:

1. **National Met Service** produces seasonal and medium-range forecasts (done monthly and weekly).
2. **District Extension Office** interprets forecasts for local context, identifies priority pests, and updates agro-meteorological bulletins (done every 10 days).
3. **IVR/SMS provider** (e.g., Viamo or similar) disseminates bulletins in local languages via automated calls and SMS (done 3× per week during growing season).
4. **Radio stations** broadcast summaries and highlight key decisions farmers face that week (done 2× per week).

5. **Extension agents** facilitate group meetings where farmers discuss forecasts, test results, and plan activities together (done every 1–2 weeks during growing season).
6. **Farmer WhatsApp groups** and informal networks amplify and contextualize messages (continuous).

This model has been implemented in parts of Senegal, Kenya, and Zambia with modest but measurable results (10–20% adoption of recommended practices, 5–15% yield increases for early adopters). The key is that all layers reinforce each other, and no single tool is expected to replace the others.[1][3][4]

5. Conclusion: The Path Forward

Digital agriculture tools are not panaceas; they are tools. Their effectiveness depends not on their sophistication but on how well they solve real problems for real farmers at prices and accessibility farmers can afford.

The gaps documented in this report—unreliable weather, generic crop advice, inaccessible interfaces, offline limitations, and language exclusion—are not insurmountable. Many are being addressed piecemeal by practitioners in the region. The Viamo 321 service in Burkina Faso shows that IVR in local languages can reach >100,000 farmers monthly. Platforms like M-Shamba and Complete Farmer demonstrate that SMS + voice can deliver more localized advice. On-farm trial networks and farmer-led research show that localized agronomic knowledge is accessible and motivating.

What is missing is systematic integration: combining what works (extension agents, local radio, farmer groups) with what is new (AI disease diagnosis, satellite rainfall data, market price information) in a design that respects farmer knowledge, works on the phones and connectivity farmers actually have, and is delivered in languages farmers speak.

For Burkina Faso and West Africa specifically, priorities are:

1. Strengthen agro-meteorological content in local languages through IVR and radio, with monthly farmer feedback on forecast accuracy.
2. Conduct farmer-validated agronomy trials for key crops in each agroecological zone, generating rules-based advisory that extension agents and digital tools can reference.
3. Develop affordable, crowdsourced disease and pest diagnostics using farmer photography and expert validation, tailored to Sahel crops.
4. Invest in offline-first, SMS-enabled platforms that work on feature phones and basic smartphones, avoiding app-only designs.
5. Establish a regional digital agriculture coordination mechanism (ECOWAS or national governments) to share tools, standards, and localized content across countries, avoiding reinvention.

The opportunity is real. Digital tools, designed with farmers rather than for them, and integrated with strong extension systems and farmer networks, can improve yields, reduce risk, and strengthen resilience in the face of climate change. The gap between ambition and reality can close—but only if tool designers listen, farmers test and feedback, and institutions commit to the longer journey of implementation, learning, and adaptation.

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